

# RYAN E. BARSZCZ

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## SUMMARY

Software Engineer with experience building full-stack applications, real-time data systems, and cloud infrastructure. Built an end-to-end vehicle telemetry platform integrating OBD-II hardware, Python, FastAPI, Next.js, PostgreSQL, and AWS. Experience with containerized deployments, asynchronous processing, CI/CD, and infrastructure as code using Docker, GitHub Actions, and Terraform.

## SKILLS

- **Languages:** JavaScript, Python, TypeScript, C++, Java, C
- **Frontend:** React, Next.js, Tailwind CSS, GSAP, React Native, Flutter
- **AWS:** ECS, Fargate, ECR, SQS, ALB, IAM, Secrets Manager, CloudWatch, S3
- **Backend & Data:** FastAPI, Node.js, Express, SQLAlchemy, Prisma, PostgreSQL, MongoDB, WebSockets
- **Cloud & DevOps:** Docker, Docker Compose, GitHub Actions, Vercel, Neon, Firebase, PyInstaller, CI/CD, Terraform
- **Testing & Tools:** Git, Postman, pytest, FastAPI TestClient, CMS, Figma, Photoshop
- **Certifications:** Google Generative AI Fundamentals (2023)

## EDUCATION

### University of Michigan–Ann Arbor

*B.S., Computer Science*

**Jun 2025**

## PROJECT EXPERIENCE

### DriveIQ - Vehicle Telemetry Platform <https://telemetry.ryanbarszcz.com/>

**Jun 2026 - Present**

- Built an end-to-end OBD-II vehicle telemetry platform using Next.js, TypeScript, FastAPI, Python, and PostgreSQL to capture, visualize, and analyze live and historical vehicle data.
- Developed and packaged a Windows Python logger that interfaces with USB OBD-II hardware, samples configurable vehicle metrics, streams live telemetry over WebSockets, and persists completed sessions as CSV data.
- Designed an asynchronous AWS telemetry pipeline using S3, SQS, and containerized background workers to decouple telemetry storage and processing from API requests.
- Deployed containerized FastAPI and worker services on AWS ECS/Fargate with ECR, an Application Load Balancer, IAM, Secrets Manager, CloudWatch logging, and CPU-based autoscaling.
- Implemented infrastructure as code with Terraform and automated CI/CD with GitHub Actions, including remote versioned S3 state, OIDC-based AWS authentication, 25+ automated API/integration tests, Docker builds, and Terraform validation/planning on pull requests.

### SetupsRus - Sim Racing Setup Sharing Platform <https://www.setupsrus.com/>

**Jun 2026**

- Built a full-stack sim racing setup-sharing platform using Next.js, TypeScript, Node.js, Express, Prisma, and PostgreSQL, enabling users to upload, browse, and download vehicle setup files.
- Developed secure authentication and ownership-based permissions with Clerk, supporting user profiles, setup management, and voting functionality.
- Integrated AWS S3 object storage and presigned URL generation for secure file uploads and downloads, enabling scalable cloud-based asset management.
- Designed and implemented RESTful APIs and relational database schemas for games, tracks, vehicles, tags, and user-generated content, supporting advanced filtering, search, and content discovery.

### Alert Arrival - Best Automotive Hackathon Winner

**Mar 2024**

#### *GrizzHacks 6*

- Prototyped a driver fatigue detection system using OpenCV eye-aspect-ratio analysis to assess drowsiness in real time. Built a Flutter UI to visualize telemetry and streamed live alerts through Firebase. Awarded Best Automotive Hack among 40+ teams for demonstrating a high-impact safety-focused mobility concept.

### Liberty Athletic Club Website

**Sep 2025 - Oct 2025**

#### *Liberty Athletic Club*

- Designed and developed a modern website for Liberty Athletic Club using Next.js, React, Tailwind CSS, and GSAP, creating a polished, responsive experience for an organization serving over 7,000 members.

## EXPERIENCE

### Liberty Athletic Club

**Feb 2023 - Present**

#### *Aquatics Director Assistant / Club Manager*

- Oversee daily operations of the high-traffic athletic facility, coordinating staff with scheduling software and resolving issues in real time, which improves member wait times and keeps the facility running smoothly.
- Lead on-shift safety coordination, incident response, and staff communication.
- Recognized as Employee of the Month for leadership and reliability 5+ times.